

DocTask

Evidence-Grounded Project Intelligence System

WHAT I BUILT

DocTask transforms scattered project documents—plans, sprint updates, milestone notes, and follow-ups—into structured, traceable, and reviewable project knowledge. The system treats documents as the **source of truth**, extracting structured facts, detecting contradictions across sources, and preserving complete evidence chains so that every conclusion in the final report can be traced back to the original material.

The core workflow is intentionally conservative: **Document Ingestion** → **Parsing** → **Chunking** → **Embedding** → **Retrieval** → **Fact Extraction** → **Normalization** → **Conflict Detection** → **Draft Generation** → **Human Review**. At each step, the system prioritizes evidence over inference, structure over summarization, and human judgment over silent automation.

CORE ENGINEERING PHILOSOPHY

The system is built around a single governing principle: **the LLM proposes, the evidence grounds, the database preserves, the conflict layer challenges, and the human decides**. This principle drives every architectural decision:

Evidence-First Retrieval		Documents are deterministically chunked and indexed using local semantic embeddings (<i>all-MiniLM-L6-v2</i>). Queries retrieve only the most relevant evidence via PostgreSQL/pgvector, avoiding the cost and noise of sending the entire corpus to the LLM.
Structured Extraction	Fact	A Groq-backed extraction agent returns strict JSON facts—subject, type, value, temporal status, source document, and supporting evidence. Facts without sufficient evidence are rejected outright.
Explicit Detection	Conflict	Contradictions—such as a milestone due date appearing as both <i>2026-08-20</i> and <i>2026-08-29</i> across sources—are surfaced for human review rather than silently merged. The system distinguishes contradictions from historical changes and ambiguity.
Incremental Updates		When new documents arrive, only affected facts are re-extracted and compared. Unaffected project knowledge remains intact, preserving previously accepted decisions and minimizing unnecessary reprocessing.

Complete Auditability	Every change is recorded with old value, new value, source document, review status, and timestamp—creating an immutable ledger of how project knowledge evolved over time.
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ARCHITECTURE & DURABILITY

The backend is built with **FastAPI** and **LangGraph**, persisting runs, documents, and stage outputs in **PostgreSQL with pgvector**. The workflow supports crash/resume (completed stages persist; interrupted runs resume from the incomplete stage), concurrent run isolation (independent runs maintain isolated retrieval contexts, facts, conflicts, and review state), and a human review gate that prevents finalization until conflicts are explicitly resolved.

Additional capabilities include post-run **Examine** validation against user-supplied rules, an **MCP interface** over Streamable HTTP for external clients, and a **watch folder** for automatic ingestion of UTF-8 text documents. A React + Vite dashboard provides project navigation, run status, workflow stage visibility, human review controls, and evidence inspection.

ENGINEERING TRADE-OFFS

I prioritized **evidence traceability** over unconstrained generation, **deterministic processing** where possible, **structured facts** over free-form summaries, and **explicit conflicts** over silent resolution. The system is scoped as a robust local implementation with 28 passing deterministic behavior tests—covering conflict resolution, incremental updates, crash/resume, concurrent isolation, prompt-injection defense, metrics accounting, and provider failure handling—rather than a claim of fully distributed production scale.

HONEST LIMITATIONS

Semantic retrieval quality depends on the embedding model and indexed evidence. LLM extraction can still omit information, so all facts retain source evidence and require validation. The watch folder is currently limited to UTF-8 `.txt` files. MCP end-to-end execution depends on local provider availability. Token and cost values depend on provider responses and environment configuration.

Core Design Principle: *The LLM proposes. The evidence grounds. The database preserves. The conflict layer challenges. The human decides.*